MSC 201 **42032**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A supplied normal distribution of sample means gives cumulative probabilities 0.15 at 8 and 0.85 at 12. What is the probability between 8 and 12?	A. 0.85. B. 0.70. C. 1.00. D. 0.15. 10 pts
2. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 0.05 percentage points. B. 12.5 percentage points. C. 5 percentage points. D. 45 percentage points. 10 pts
3. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 6. B. 2. C. 12. D. 72. 10 pts
4. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 17. B. 15. C. 16. D. 256. 10 pts
5. A sample grows from 50 to 51 observations. Must its mean move closer to the population mean?	A. No; individual additions can move it either way. B. Yes; the law of large numbers guarantees monotone improvement. C. Yes; every increase improves the estimate. D. No; larger samples always worsen estimates. 10 pts
6. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Both quadruple. B. Expected sum stays fixed; SD halves. C. Expected sum quadruples; SD doubles. D. Both halve. 10 pts
7. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. Yes; n=100 guarantees every condition. B. No; proportions cannot be estimated at all. C. Yes; failures are numerous. D. No; the observed success count is too sparse. 10 pts

Test continues on the other side of the sheet.

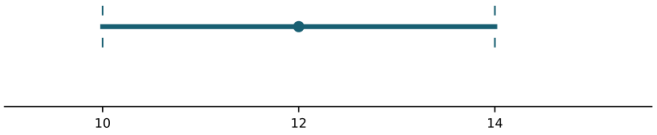
MSC 201 **42032**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A shipment contains 36 packages. Which variable answers the question of average mass per package in that shipment?	A. The largest mass. B. The total mass without dividing. C. The sample mean. D. A single randomly chosen mass. 10 pts
9. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. Both widths must be identical. B. The 99% interval has a different sample mean. C. The 99% interval is wider but has the same center. D. The 90% interval is wider. 10 pts
10. Sample mean is 12 and margin of error is 2. What interval results? 	A. 12 to 14. B. 6 to 24. C. 2 to 12. D. 10 to 14. 10 pts

42032

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

C.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

C.

10 pts

10.

D.

10 pts

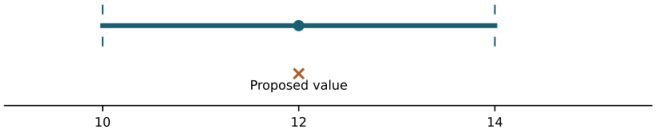
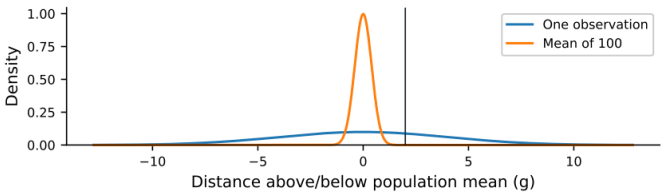
MSC 201 **63591**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A mean interval is 10 to 14. How should a proposed mean of 12 be described? 	A. Impossible. B. The only compatible value. C. Compatible with the interval, without being proved exact. D. Proved to be the exact population mean. 10 pts
2. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Thirty is too large for any sample. B. Only populations with mean zero qualify. C. Sample means can never approach normality. D. Skewness and extreme observations affect how large n needs to be. 10 pts
3. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. About 95% of repeated sample means are at most 14. B. The unknown mean has become 14. C. 95% of all individuals must be at most 14. D. Every sample mean is 14. 10 pts
4. For a normal population, compare one mass exceeding mean+2 g with the mean of 100 independent masses exceeding the same cutoff. Which is more likely? 	A. Both have probability zero. B. They are equally likely. C. The 100-mass mean event. D. The individual mass event. 10 pts
5. An open voluntary poll has 50,000 replies and a very small calculated margin. What remains uncertain?	A. Whether the calculated margin must be negative. B. Whether a proportion is numerical. C. Whether its respondents represent the target population. D. Whether 50,000 exceeds 30. 10 pts

Test continues on the other side of the sheet.

MSC 201 **63591**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. A sum event and its equivalent mean event use correctly matched centers and SDs. How do their z-scores compare?	A. The sum z-score is n times larger. B. The mean z-score is always zero. C. They are equal. D. Their signs must be opposite. 10 pts
7. Which t distribution has heavier tails?	A. Both have no tails. B. They have equal tails. C. df=100. D. df=4. 10 pts
8. A researcher takes 1,000 samples of 25 packages and plots each sample mean. What does one plotted value represent?	A. The mean of one 25-package sample. B. A population parameter that changes on each sample. C. The mass of one package. D. The mean of all 25,000 packages combined. 10 pts
9. Which question targets a population proportion?	A. What is the mean duration of jobs? B. What is the median number of pages? C. What is the SD of file sizes? D. What fraction of jobs finish on time? 10 pts
10. For known population SD and fixed confidence, n quadruples. What happens to the margin of error?	A. It quarters. B. It halves. C. It doubles. D. It stays unchanged. 10 pts

63591

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

C.

10 pts

TEST PAGE 2

6.

C.

10 pts

7.

D.

10 pts

8. A.

10 pts

9.

D.

10 pts

10.

B.

10 pts

MSC 201 **83385**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Individual masses have SD 20 g; means of 25 independent masses have SD 4 g. Which spread applies to one package?	A. 0.8 g. B. 4 g. C. 16 g. D. 20 g. 10 pts
2. Twenty items have sample mean mass 5 g. What is their total mass?	A. 0.25 g. B. 25 g. C. 5 g. D. 100 g. 10 pts
3. A proportion estimate is 0.40 with margin 0.05. What interval is this in percentages?	A. 40% to 50%. B. 5% to 40%. C. 39.95% to 40.05%. D. 35% to 45%. 10 pts
4. Increasing n from 100 to 400 halves SE. Would adding the same 300 observations to any starting n always halve SE?	A. Yes; an extra 300 always halves it. B. No; SE never changes. C. No; the ratio of sample sizes matters. D. Yes; SE depends only on added observations. 10 pts
5. A small random independent sample comes from a normal population. Is n<30 alone a reason to reject a t interval?	A. Yes; normality is irrelevant below 30. B. No; small normal samples can support the t method. C. Yes; t requires at least 30 in every case. D. No; it must use a proportion interval. 10 pts
6. Which continuous event uses the appropriate continuity correction for X>=45?	A. Y>44.5. B. Y<44.5. C. Y=45. D. Y>45.5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **83385**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. No; it estimates the population mean. B. Yes; the mean and individuals have identical spread. C. Yes; that is the meaning of any confidence interval. D. No; it instead guarantees every duration lies there. 10 pts
8. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. It is approximation error outside the true support. B. The true sample mean can actually be below every observation. C. The observations must include negative waits. D. The normal model is exact at every tail. 10 pts
9. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. It decreases. B. It increases. C. It stays unchanged. D. The sample proportion becomes the confidence level. 10 pts
10. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. The 99% interval has a different sample mean. B. The 90% interval is wider. C. Both widths must be identical. D. The 99% interval is wider but has the same center. 10 pts

83385

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6. A.

10 pts

TEST PAGE 2

7. A.

10 pts

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **97261**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For n=20, a total exceeding 240 is equivalent to which mean event?	A. Mean exceeding 12. B. Mean below 12. C. Mean exceeding 240. D. Mean exceeding 20. 10 pts
2. A mean interval is 10 to 14 minutes. Which report matches its target?	A. Every future duration will be between 10 and 14. B. The procedure estimates the population mean duration between 10 and 14 minutes. C. The median is guaranteed to equal 12. D. Exactly 95 of the next 100 durations will lie there. 10 pts
3. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether the mean is a whole number. B. Whether every observation equals the mean. C. Whether the interval includes zero. D. Sampling quality and independence. 10 pts
4. For independent observations with finite variance, compare repeated means at n=1 and n=100. What is generally expected?	A. Every size-100 mean is identical. B. The n=1 histogram must be normal. C. The population itself becomes less spread. D. Means at n=100 are less spread and often more nearly normal. 10 pts
5. A question asks whether the combined duration of 40 tasks exceeds one hour. Which variable should be modeled?	A. The median alone. B. One individual duration. C. The count of tasks alone. D. The sum of 40 durations. 10 pts
6. A researcher takes 1,000 samples of 25 packages and plots each sample mean. What does one plotted value represent?	A. The mean of all 25,000 packages combined. B. A population parameter that changes on each sample. C. The mass of one package. D. The mean of one 25-package sample. 10 pts
7. The success proportion is 0.35. What is the failure proportion?	A. 0.65. B. 0.35. C. 0.175. D. 1.35. 10 pts

Test continues on the other side of the sheet.

MSC 201 **97261**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 6. B. 12. C. 2. D. 72. 10 pts
9. Everyone in a few selected clubs is surveyed. What challenges treating all responses as independent?	A. Club members cannot have binary responses. B. The sample proportion cannot be computed. C. Randomly selecting clubs guarantees individual independence. D. Members of the same club may have correlated responses. 10 pts
10. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It becomes the population SD. B. It stays unchanged. C. It increases because there are more observations. D. It decreases as SE and the critical t value decrease. 10 pts

97261

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. D.

10 pts

4. D.

10 pts

5. D.

10 pts

6. D.

10 pts

7. A.

10 pts

TEST PAGE 2

8. C.

10 pts

9. D.

10 pts

10. D.

10 pts

MSC 201 **63267**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. Yes; any other count proves bias. B. No; all 100 must cover it. C. Yes; the confidence level fixes that count. D. No; the coverage count itself can vary. 10 pts
2. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. No; n must exceed 100. B. Yes; plus-four has no stated conditions. C. No; the confidence level is below the stated threshold. D. Yes; n alone is sufficient. 10 pts
3. 40 of 100 sampled jobs succeed. What is the sample success proportion?	A. 0.60. B. 40. C. 0.40. D. 2.5. 10 pts
4. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 15. C. 17. D. 16. 10 pts
5. Only successful downloads are sampled to estimate duration for all attempts. Does a small SE remove that selection problem?	A. No; precision does not correct selection bias. B. Yes; a normal curve proves unbiased selection. C. No; SE must equal the mean. D. Yes; enough successful downloads represent failures automatically. 10 pts
6. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether every observation equals the mean. B. Sampling quality and independence. C. Whether the mean is a whole number. D. Whether the interval includes zero. 10 pts
7. Population SD doubles while sample size stays fixed. What happens to SE?	A. It quadruples. B. It stays unchanged. C. It doubles. D. It halves. 10 pts

Test continues on the other side of the sheet.

MSC 201 **63267**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It becomes zero. B. It stays at the population mean. C. It quadruples. D. It halves. 10 pts
9. Which interval approximates the discrete event X=45?	A. Y>45.5. B. Y=45 only. C. 44.5<Y<45.5. D. Y<44.5. 10 pts
10. A fixed 5-kg container is added to a shipment of random item masses. What changes?	A. Expected total rises by 5 kg; total SD is unchanged. B. Total SD doubles. C. Expected total stays fixed. D. Both mean and SD rise by 5 kg. 10 pts

63267

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **76726**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. No; the coverage count itself can vary. B. Yes; any other count proves bias. C. No; all 100 must cover it. D. Yes; the confidence level fixes that count. 10 pts
2. The course's binomial normal approximation check requires np>5 and n(1-p)>5. For n=100, p=0.02, which judgment fits?	A. It fails because expected successes are only 2. B. It passes because n exceeds 30. C. It passes because p is positive. D. It fails because expected failures are 98. 10 pts
3. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It decreases. B. It becomes zero. C. It stays unchanged. D. It increases. 10 pts
4. A population is strongly right-skewed and n=5. What is the best judgment about a normal model for the mean?	A. It must be left-skewed. B. It is exact for every skewed population. C. It is guaranteed by the existence of a mean. D. It is not automatically justified. 10 pts
5. Which quantity describes how sample means vary from sample to sample?	A. Standard error. B. The population mean. C. The sample size. D. The range of all individuals alone. 10 pts
6. To halve SE, how must an independent sample's size change?	A. Add 4 regardless of its current size. B. Multiply n by 2. C. Multiply n by 4. D. Divide n by 2. 10 pts
7. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 0.25. B. 90. C. 120. D. 30. 10 pts

Test continues on the other side of the sheet.

MSC 201 **76726**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A one-sample t interval. B. A Poisson count probability. C. A known-SD z interval treating s as known population SD. D. A proportion interval regardless of variable type. 10 pts
9. The course's ordinary normal-approximation interval check requires more than 5 observed successes and more than 5 failures. Which sample passes?	A. 5 successes and 95 failures. B. 20 successes and 0 failures. C. 8 successes and 12 failures. D. 2 successes and 98 failures. 10 pts
10. Individual mean mass is 3 g. What is the expected total for 10 items?	A. 30 g. B. 3 g. C. 10 g. D. Exactly 30 g in every shipment. 10 pts

76726

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. D.

10 pts

4. D.

10 pts

5. A.

10 pts

6. C.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9. C.

10 pts

10. A.

10 pts

MSC 201 **82883**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. In 1,000 repetitions, each sample contains 25 independent observations. What n belongs in SD/sqrt(n) for the mean?	A. 25. B. 975. C. 1,000. D. 25,000. 10 pts
2. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. The unknown mean has become 14. B. 95% of all individuals must be at most 14. C. Every sample mean is 14. D. About 95% of repeated sample means are at most 14. 10 pts
3. The course's ordinary normal-approximation interval check requires more than 5 observed successes and more than 5 failures. Which sample passes?	A. 20 successes and 0 failures. B. 8 successes and 12 failures. C. 5 successes and 95 failures. D. 2 successes and 98 failures. 10 pts
4. A population is strongly right-skewed and n=5. What is the best judgment about a normal model for the mean?	A. It is exact for every skewed population. B. It is guaranteed by the existence of a mean. C. It must be left-skewed. D. It is not automatically justified. 10 pts
5. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Sampling quality and independence. B. Whether the interval includes zero. C. Whether the mean is a whole number. D. Whether every observation equals the mean. 10 pts
6. A sample of 6 has strong skewness and an extreme outlier. What is sensible about a t interval?	A. It is automatically reliable because t has heavy tails. B. Six observations guarantee a normal sampling distribution. C. Its reliability deserves caution. D. The outlier must be removed without investigation. 10 pts

Test continues on the other side of the sheet.

MSC 201 **82883**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A sum event and its equivalent mean event use correctly matched centers and SDs. How do their z-scores compare?	A. They are equal. B. The sum z-score is n times larger. C. The mean z-score is always zero. D. Their signs must be opposite. 10 pts
8. Sample means are centered at 40 with SE 2. Which mean is 1.5 SE above center?	A. 37. B. 43. C. 41.5. D. 42. 10 pts
9. Which question targets a population proportion?	A. What is the SD of file sizes? B. What is the median number of pages? C. What is the mean duration of jobs? D. What fraction of jobs finish on time? 10 pts
10. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The mean occupies 95% of the interval. B. The fixed mean changes to fit it with probability 95%. C. It either contains that mean or does not; 95% describes the procedure. D. The endpoints must equal the smallest and largest observation. 10 pts

82883

TEST PAGE 1

1. A.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

C.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **17693**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. The sample proportion becomes the confidence level. B. It increases. C. It decreases. D. It stays unchanged. 10 pts
2. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Skewness and extreme observations affect how large n needs to be. B. Sample means can never approach normality. C. Only populations with mean zero qualify. D. Thirty is too large for any sample. 10 pts
3. Twenty items have sample mean mass 5 g. What is their total mass?	A. 5 g. B. 25 g. C. 0.25 g. D. 100 g. 10 pts
4. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. Yes; all sample proportions equal 0.95. B. No; it means 95 people were surveyed. C. Yes; confidence equals the use rate. D. No; 95% describes the interval procedure's coverage. 10 pts
5. A shipment contains 36 packages. Which variable answers the question of average mass per package in that shipment?	A. The sample mean. B. A single randomly chosen mass. C. The total mass without dividing. D. The largest mass. 10 pts
6. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 72. B. 2. C. 12. D. 6. 10 pts
7. A report gives sample SD=6 minutes and SE=1 minute. Which describes variation among individual observed times?	A. The confidence level. B. 7 minutes. C. 1 minute. D. 6 minutes. 10 pts

Test continues on the other side of the sheet.

MSC 201 **17693**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It increases. B. It decreases. C. It stays unchanged. D. It becomes zero. 10 pts
9. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. The unknown mean has become 14. B. About 95% of repeated sample means are at most 14. C. 95% of all individuals must be at most 14. D. Every sample mean is 14. 10 pts
10. Why report a confidence interval instead of only a sample mean?	A. It lists every individual observation. B. It removes sampling bias. C. It guarantees the true mean is the midpoint. D. It expresses uncertainty in estimating the population mean. 10 pts

17693

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

B.

10 pts

7.

D.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

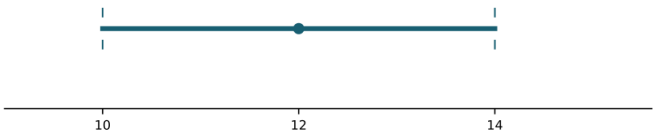
MSC 201 **71574**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Independent observations come from an exactly normal population. Is n>=30 necessary for their mean to be normal?	A. No; their mean is normal for any positive sample size. B. No; the mean instead becomes uniform. C. Yes; individual normality disappears in small samples. D. Yes; normal means are impossible below 30. 10 pts
2. A researcher takes 1,000 samples of 25 packages and plots each sample mean. What does one plotted value represent?	A. A population parameter that changes on each sample. B. The mean of all 25,000 packages combined. C. The mass of one package. D. The mean of one 25-package sample. 10 pts
3. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 15. B. 256. C. 17. D. 16. 10 pts
4. Sample mean is 12 and margin of error is 2. What interval results? 	A. 10 to 14. B. 12 to 14. C. 2 to 12. D. 6 to 24. 10 pts
5. Why use planning proportion 0.5 when no prior estimate is available?	A. It maximizes p(1-p), giving a conservative required sample size. B. It minimizes all margins to zero. C. It guarantees half the population has the trait. D. It makes the sample unbiased automatically. 10 pts

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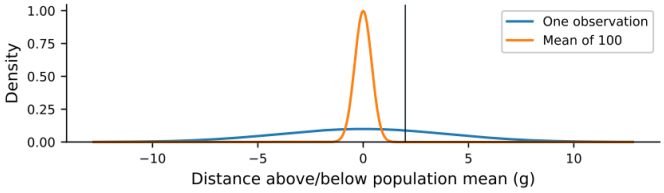
MSC 201 **71574**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. For a normal population, compare one mass exceeding mean+2 g with the mean of 100 independent masses exceeding the same cutoff. Which is more likely? 	A. Both have probability zero. B. The 100-mass mean event. C. The individual mass event. D. They are equally likely. 10 pts
7. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. The unknown mean has become 14. B. About 95% of repeated sample means are at most 14. C. 95% of all individuals must be at most 14. D. Every sample mean is 14. 10 pts
8. For n=20, a total exceeding 240 is equivalent to which mean event?	A. Mean below 12. B. Mean exceeding 12. C. Mean exceeding 240. D. Mean exceeding 20. 10 pts
9. A narrow interval uses only the fastest successful jobs to estimate the mean for all jobs. What is its main limitation?	A. Biased selection can make the interval misleading despite narrow width. B. Its narrow width guarantees representativeness. C. Adding decimal places removes selection bias. D. The mean cannot describe time. 10 pts
10. A proportion estimate is 0.40 with margin 0.05. What interval is this in percentages?	A. 40% to 50%. B. 35% to 45%. C. 39.95% to 40.05%. D. 5% to 40%. 10 pts

71574

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. A.

10 pts

4. A.

10 pts

5. A.

10 pts

TEST PAGE 2

6. C.

10 pts

7. B.

10 pts

8. B.

10 pts

9. A.

10 pts

10. B.

10 pts

MSC 201 **55735**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 72. B. 6. C. 12. D. 2. 10 pts
2. Independent observations come from an exactly normal population. Is n>=30 necessary for their mean to be normal?	A. No; the mean instead becomes uniform. B. Yes; normal means are impossible below 30. C. Yes; individual normality disappears in small samples. D. No; their mean is normal for any positive sample size. 10 pts
3. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It decreases. B. It stays unchanged. C. It increases. D. It becomes zero. 10 pts
4. A sample of 6 has strong skewness and an extreme outlier. What is sensible about a t interval?	A. It is automatically reliable because t has heavy tails. B. Its reliability deserves caution. C. The outlier must be removed without investigation. D. Six observations guarantee a normal sampling distribution. 10 pts
5. Everyone in a few selected clubs is surveyed. What challenges treating all responses as independent?	A. The sample proportion cannot be computed. B. Club members cannot have binary responses. C. Randomly selecting clubs guarantees individual independence. D. Members of the same club may have correlated responses. 10 pts
6. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. No; it estimates the population mean. B. No; it instead guarantees every duration lies there. C. Yes; that is the meaning of any confidence interval. D. Yes; the mean and individuals have identical spread. 10 pts

Test continues on the other side of the sheet.

MSC 201 **55735**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. The course's binomial normal approximation check requires np>5 and n(1-p)>5. For n=100, p=0.02, which judgment fits?	A. It fails because expected successes are only 2. B. It passes because n exceeds 30. C. It passes because p is positive. D. It fails because expected failures are 98. 10 pts
8. Which quantity describes how sample means vary from sample to sample?	A. Standard error. B. The sample size. C. The range of all individuals alone. D. The population mean. 10 pts
9. A symmetric proportion interval is 0.20 to 0.30. What is the estimate and margin?	A. Estimate 0.10; margin 0.05. B. Estimate 0.25; margin 0.05. C. Estimate 0.20; margin 0.10. D. Estimate 0.30; margin 0.25. 10 pts
10. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It predicts a nonnegative SD. B. It retains mass units. C. It uses the sample size. D. It assumes independent variation. 10 pts

55735

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6. A.

10 pts

TEST PAGE 2

7. A.

10 pts

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **34462**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. The larger sample changes the claimed mean. B. They are equally far. C. The size-100 mean. D. The size-25 mean. 10 pts
2. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It predicts a nonnegative SD. B. It assumes independent variation. C. It retains mass units. D. It uses the sample size. 10 pts
3. For a known-SD mean interval, supplied SE=0.5 and critical multiplier=2. What is the margin of error?	A. 2.5. B. 0.25. C. 1. D. 4. 10 pts
4. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. No; it means 95 people were surveyed. B. No; 95% describes the interval procedure's coverage. C. Yes; all sample proportions equal 0.95. D. Yes; confidence equals the use rate. 10 pts
5. Which continuous event uses the appropriate continuity correction for X>=45?	A. Y>44.5. B. Y<44.5. C. Y=45. D. Y>45.5. 10 pts
6. A sample of 6 has strong skewness and an extreme outlier. What is sensible about a t interval?	A. Six observations guarantee a normal sampling distribution. B. Its reliability deserves caution. C. The outlier must be removed without investigation. D. It is automatically reliable because t has heavy tails. 10 pts

Test continues on the other side of the sheet.

MSC 201 **34462**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. An open voluntary poll has 50,000 replies and a very small calculated margin. What remains uncertain?	A. Whether 50,000 exceeds 30. B. Whether a proportion is numerical. C. Whether its respondents represent the target population. D. Whether the calculated margin must be negative. 10 pts
8. A sample grows from 50 to 51 observations. Must its mean move closer to the population mean?	A. No; individual additions can move it either way. B. Yes; the law of large numbers guarantees monotone improvement. C. Yes; every increase improves the estimate. D. No; larger samples always worsen estimates. 10 pts
9. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It quadruples. B. It stays at the population mean. C. It halves. D. It becomes zero. 10 pts
10. Why report a confidence interval instead of only a sample mean?	A. It removes sampling bias. B. It lists every individual observation. C. It guarantees the true mean is the midpoint. D. It expresses uncertainty in estimating the population mean. 10 pts

34462

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

C.

10 pts

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **63786**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. SE is 4 at n=25. What is SE at n=100, with the same population SD and independence?	A. 1. B. 8. C. 2. D. 4. 10 pts
2. For a comparison holding the observed proportion and confidence fixed, n quadruples. What happens to estimated margin?	A. It doubles. B. It halves. C. It stays unchanged. D. It quarters. 10 pts
3. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 90. B. 0.25. C. 30. D. 120. 10 pts
4. The CLT supports an approximately normal distribution of large-sample means. What does it not claim?	A. Means can have approximately normal sampling distributions. B. Sampling distributions concern statistics. C. Larger independent samples can have smaller SE. D. Individual observations become normally distributed. 10 pts
5. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 17. B. 256. C. 16. D. 15. 10 pts
6. Independent sample size increases. What happens to spread in original units?	A. Both rise. B. Both fall. C. Both stay unchanged. D. SD of sums rises; SE of averages falls. 10 pts

Test continues on the other side of the sheet.

MSC 201 **63786**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The fixed mean changes to fit it with probability 95%. B. It either contains that mean or does not; 95% describes the procedure. C. The mean occupies 95% of the interval. D. The endpoints must equal the smallest and largest observation. 10 pts
8. Why approximate P(X=40) for a binomial count using a normal interval around 40 instead of a single point?	A. Every count equals its mean. B. The count 40 is impossible. C. A normal model has no area. D. A continuous model assigns zero probability to one exact point. 10 pts
9. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It becomes zero. B. It stays at the population mean. C. It halves. D. It quadruples. 10 pts
10. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It stays unchanged. B. It becomes zero. C. It increases. D. It decreases. 10 pts

63786

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **62332**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. It increases. B. It stays unchanged. C. It decreases. D. The sample proportion becomes the confidence level. 10 pts
2. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. The sample mean equals the population mean 95% of the time. B. About 95% of intervals constructed this way contain the fixed population mean in the long run. C. Every interval has 95% of its length below the mean. D. 95% of observations lie in every interval. 10 pts
3. Population SD doubles while sample size stays fixed. What happens to SE?	A. It stays unchanged. B. It doubles. C. It halves. D. It quadruples. 10 pts
4. The course's binomial normal approximation check requires $np>5$ and $n(1-p)>5$. For $n=100$, $p=0.02$, which judgment fits?	A. It passes because p is positive. B. It fails because expected failures are 98. C. It fails because expected successes are only 2. D. It passes because n exceeds 30. 10 pts
5. A small random independent sample comes from a normal population. Is $n<30$ alone a reason to reject a t interval?	A. No; small normal samples can support the t method. B. Yes; normality is irrelevant below 30. C. No; it must use a proportion interval. D. Yes; t requires at least 30 in every case. 10 pts
6. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether the mean is a whole number. B. Sampling quality and independence. C. Whether every observation equals the mean. D. Whether the interval includes zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **62332**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A sample grows from 50 to 51 observations. Must its mean move closer to the population mean?	A. No; larger samples always worsen estimates. B. No; individual additions can move it either way. C. Yes; every increase improves the estimate. D. Yes; the law of large numbers guarantees monotone improvement. 10 pts
8. A fixed 5-kg container is added to a shipment of random item masses. What changes?	A. Both mean and SD rise by 5 kg. B. Expected total stays fixed. C. Expected total rises by 5 kg; total SD is unchanged. D. Total SD doubles. 10 pts
9. A shipment contains 36 packages. Which variable answers the question of average mass per package in that shipment?	A. A single randomly chosen mass. B. The sample mean. C. The largest mass. D. The total mass without dividing. 10 pts
10. A proportion interval is 20% to 30%. Which conclusion matches the target?	A. Exactly 25% of the sample must be successful. B. Every individual has a value between 20 and 30. C. The population percentage is estimated between 20% and 30%. D. The mean duration is between 20 and 30 minutes. 10 pts

62332

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. C.

10 pts

5. A.

10 pts

6. B.

10 pts

TEST PAGE 2

7. B.

10 pts

8. C.

10 pts

9. B.

10 pts

10. C.

10 pts

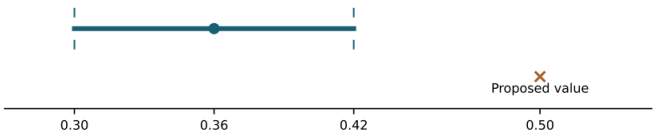
MSC 201 **80271**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It decreases as SE and the critical t value decrease. B. It becomes the population SD. C. It stays unchanged. D. It increases because there are more observations. 10 pts
2. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. Yes; n=100 guarantees every condition. B. No; the observed success count is too sparse. C. Yes; failures are numerous. D. No; proportions cannot be estimated at all. 10 pts
3. Which interval approximates the discrete event X=45?	A. Y>45.5. B. Y=45 only. C. 44.5<Y<45.5. D. Y<44.5. 10 pts
4. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Multiply n by 4. B. Halve n. C. Multiply n by 2. D. Add 1 to n. 10 pts
5. A population proportion interval is 0.30 to 0.42. How should a proposed proportion 0.50 be described? 	A. The required sample estimate. B. Proved impossible under every future sample. C. Compatible because 0.50 is a valid probability. D. Outside the interval and incompatible with this interval estimate. 10 pts
6. In 1,000 repetitions, each sample contains 25 independent observations. What n belongs in SD/sqrt(n) for the mean?	A. 975. B. 25,000. C. 1,000. D. 25. 10 pts

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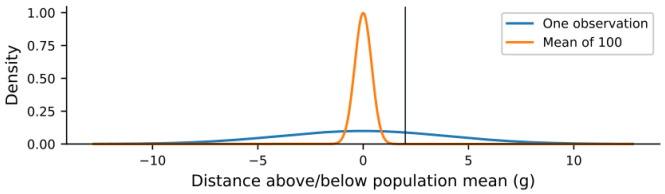
MSC 201 **80271**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Fifty readings taken seconds apart from one fluctuating machine are treated as independent. What should be checked?	A. Dependence between nearby readings. B. Whether the mean can be calculated. C. Whether readings have units. D. Whether 50 is an integer. 10 pts
8. For a normal population, compare one mass exceeding mean+2 g with the mean of 100 independent masses exceeding the same cutoff. Which is more likely? 	A. The 100-mass mean event. B. The individual mass event. C. Both have probability zero. D. They are equally likely. 10 pts
9. A symmetric confidence interval is 8 to 12. What point estimate is at its center?	A. 8. B. 10. C. 4. D. 12. 10 pts
10. For n=20, a total exceeding 240 is equivalent to which mean event?	A. Mean exceeding 20. B. Mean exceeding 12. C. Mean below 12. D. Mean exceeding 240. 10 pts

80271

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. C.

10 pts

4. A.

10 pts

5. D.

10 pts

6. D.

10 pts

TEST PAGE 2

7. A.

10 pts

8. B.

10 pts

9. B.

10 pts

10. B.

10 pts

MSC 201 **45365**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A supplied normal distribution of sample means gives cumulative probabilities 0.15 at 8 and 0.85 at 12. What is the probability between 8 and 12?	A. 0.15. B. 0.70. C. 1.00. D. 0.85. 10 pts
2. A proportion interval is 20% to 30%. Which conclusion matches the target?	A. The population percentage is estimated between 20% and 30%. B. Every individual has a value between 20 and 30. C. The mean duration is between 20 and 30 minutes. D. Exactly 25% of the sample must be successful. 10 pts
3. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. It is approximation error outside the true support. B. The true sample mean can actually be below every observation. C. The normal model is exact at every tail. D. The observations must include negative waits. 10 pts
4. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A proportion interval regardless of variable type. B. A Poisson count probability. C. A known-SD z interval treating s as known population SD. D. A one-sample t interval. 10 pts
5. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Multiply n by 4. B. Multiply n by 2. C. Halve n. D. Add 1 to n. 10 pts
6. Independent sample size quadruples with population SD fixed. What happens to SE of the mean?	A. It halves. B. It quarters. C. It stays unchanged. D. It doubles. 10 pts

Test continues on the other side of the sheet.

MSC 201 **45365**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. No; all 100 must cover it. B. Yes; the confidence level fixes that count. C. Yes; any other count proves bias. D. No; the coverage count itself can vary. 10 pts
8. Individual masses have SD 20 g; means of 25 independent masses have SD 4 g. Which spread applies to one package?	A. 0.8 g. B. 4 g. C. 16 g. D. 20 g. 10 pts
9. The course's ordinary normal-approximation interval check requires more than 5 observed successes and more than 5 failures. Which sample passes?	A. 20 successes and 0 failures. B. 2 successes and 98 failures. C. 8 successes and 12 failures. D. 5 successes and 95 failures. 10 pts
10. Independent sample size increases. What happens to spread in original units?	A. Both stay unchanged. B. SD of sums rises; SE of averages falls. C. Both fall. D. Both rise. 10 pts

45365

TEST PAGE 1

1. B.

10 pts

2. A.

10 pts

3. A.

10 pts

4. D.

10 pts

5. A.

10 pts

6. A.

10 pts

TEST PAGE 2

7. D.

10 pts

8. D.

10 pts

9. C.

10 pts

10. B.

10 pts

MSC 201 **28952**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A small random independent sample comes from a normal population. Is n<30 alone a reason to reject a t interval?	A. Yes; t requires at least 30 in every case. B. No; it must use a proportion interval. C. No; small normal samples can support the t method. D. Yes; normality is irrelevant below 30. 10 pts
2. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. Yes; n alone is sufficient. B. No; n must exceed 100. C. No; the confidence level is below the stated threshold. D. Yes; plus-four has no stated conditions. 10 pts
3. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It becomes zero. B. It stays unchanged. C. It increases. D. It decreases. 10 pts
4. Increasing n from 100 to 400 halves SE. Would adding the same 300 observations to any starting n always halve SE?	A. No; SE never changes. B. No; the ratio of sample sizes matters. C. Yes; SE depends only on added observations. D. Yes; an extra 300 always halves it. 10 pts
5. Why report a confidence interval instead of only a sample mean?	A. It lists every individual observation. B. It removes sampling bias. C. It guarantees the true mean is the midpoint. D. It expresses uncertainty in estimating the population mean. 10 pts
6. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Both halve. B. Both quadruple. C. Expected sum quadruples; SD doubles. D. Expected sum stays fixed; SD halves. 10 pts
7. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It stays at the population mean. B. It becomes zero. C. It quadruples. D. It halves. 10 pts

Test continues on the other side of the sheet.

MSC 201 **28952**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Which interval approximates the discrete event X=45?	A. 44.5<Y<45.5. B. Y=45 only. C. Y<44.5. D. Y>45.5. 10 pts
9. A proportion estimate is 0.40 with margin 0.05. What interval is this in percentages?	A. 39.95% to 40.05%. B. 35% to 45%. C. 40% to 50%. D. 5% to 40%. 10 pts
10. Fifty readings taken seconds apart from one fluctuating machine are treated as independent. What should be checked?	A. Whether readings have units. B. Whether 50 is an integer. C. Dependence between nearby readings. D. Whether the mean can be calculated. 10 pts

28952

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

C.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **21462**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Which continuous event uses the appropriate continuity correction for X>=45?	A. Y=45. B. Y<44.5. C. Y>44.5. D. Y>45.5. 10 pts
2. Which t distribution has heavier tails?	A. They have equal tails. B. df=4. C. Both have no tails. D. df=100. 10 pts
3. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Sample means can never approach normality. B. Skewness and extreme observations affect how large n needs to be. C. Thirty is too large for any sample. D. Only populations with mean zero qualify. 10 pts
4. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It becomes zero. B. It stays at the population mean. C. It quadruples. D. It halves. 10 pts
5. Increasing n from 100 to 400 halves SE. Would adding the same 300 observations to any starting n always halve SE?	A. Yes; SE depends only on added observations. B. No; the ratio of sample sizes matters. C. No; SE never changes. D. Yes; an extra 300 always halves it. 10 pts
6. Twenty items have sample mean mass 5 g. What is their total mass?	A. 25 g. B. 5 g. C. 100 g. D. 0.25 g. 10 pts
7. For a comparison holding the observed proportion and confidence fixed, n quadruples. What happens to estimated margin?	A. It halves. B. It quarters. C. It doubles. D. It stays unchanged. 10 pts

Test continues on the other side of the sheet.

MSC 201 **21462**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. About 95% of intervals constructed this way contain the fixed population mean in the long run. B. Every interval has 95% of its length below the mean. C. 95% of observations lie in every interval. D. The sample mean equals the population mean 95% of the time. 10 pts
9. A narrow interval uses only the fastest successful jobs to estimate the mean for all jobs. What is its main limitation?	A. Adding decimal places removes selection bias. B. Its narrow width guarantees representativeness. C. Biased selection can make the interval misleading despite narrow width. D. The mean cannot describe time. 10 pts
10. The success proportion is 0.35. What is the failure proportion?	A. 0.35. B. 0.175. C. 0.65. D. 1.35. 10 pts

21462

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

C.

10 pts

10.

C.

10 pts

MSC 201 **76524**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For a comparison holding the observed proportion and confidence fixed, n quadruples. What happens to estimated margin?	A. It quarters. B. It halves. C. It stays unchanged. D. It doubles. 10 pts
2. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. No; 95% describes the interval procedure's coverage. B. Yes; all sample proportions equal 0.95. C. Yes; confidence equals the use rate. D. No; it means 95 people were surveyed. 10 pts
3. Which interval approximates the discrete event X=45?	A. Y<44.5. B. Y>45.5. C. 44.5<Y<45.5. D. Y=45 only. 10 pts
4. Individual mean mass is 3 g. What is the expected total for 10 items?	A. Exactly 30 g in every shipment. B. 30 g. C. 3 g. D. 10 g. 10 pts
5. Why does a t interval use a larger critical multiplier than z at small degrees of freedom?	A. Estimating SD adds uncertainty reflected in heavier tails. B. The confidence level is automatically 100%. C. The population mean is random in the formula. D. The observations must all be larger. 10 pts
6. A symmetric 90% confidence construction leaves what fraction in each tail?	A. 0.05. B. 0.90. C. 0.45. D. 0.10. 10 pts
7. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 4. B. 12. C. 2. D. 0.5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **76524**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. Yes; the mean and individuals have identical spread. B. No; it instead guarantees every duration lies there. C. Yes; that is the meaning of any confidence interval. D. No; it estimates the population mean. 10 pts
9. Only successful downloads are sampled to estimate duration for all attempts. Does a small SE remove that selection problem?	A. No; precision does not correct selection bias. B. Yes; enough successful downloads represent failures automatically. C. Yes; a normal curve proves unbiased selection. D. No; SE must equal the mean. 10 pts
10. In 1,000 repetitions, each sample contains 25 independent observations. What n belongs in SD/sqrt(n) for the mean?	A. 25,000. B. 1,000. C. 975. D. 25. 10 pts

76524

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5. A.

10 pts

6. A.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **45813**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The population mean is unknown. Why are repeated-sample models useful for estimation?	A. They remove the need for representative sampling. B. They guarantee every sample equals the population. C. They make the population mean change randomly every day. D. They describe how far a sample mean might vary around that population mean. 10 pts
2. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 104. B. 200. C. 102. D. 108. 10 pts
3. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. Every interval has 95% of its length below the mean. B. 95% of observations lie in every interval. C. The sample mean equals the population mean 95% of the time. D. About 95% of intervals constructed this way contain the fixed population mean in the long run. 10 pts
4. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. It stays unchanged. B. It increases. C. It decreases. D. The sample proportion becomes the confidence level. 10 pts
5. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 0.5. B. 2. C. 4. D. 12. 10 pts
6. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. 95% of all individuals must be at most 14. B. About 95% of repeated sample means are at most 14. C. The unknown mean has become 14. D. Every sample mean is 14. 10 pts

Test continues on the other side of the sheet.

MSC 201 **45813**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Which question targets a population proportion?	A. What is the median number of pages? B. What fraction of jobs finish on time? C. What is the SD of file sizes? D. What is the mean duration of jobs? 10 pts
8. Fifty readings taken seconds apart from one fluctuating machine are treated as independent. What should be checked?	A. Whether the mean can be calculated. B. Whether readings have units. C. Dependence between nearby readings. D. Whether 50 is an integer. 10 pts
9. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It becomes the population SD. B. It decreases as SE and the critical t value decrease. C. It increases because there are more observations. D. It stays unchanged. 10 pts
10. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether the mean is a whole number. B. Whether every observation equals the mean. C. Sampling quality and independence. D. Whether the interval includes zero. 10 pts

45813

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

C.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **32475**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A sample-size formula returns 62.2 observations. Which planned integer size meets that minimum?	A. 62. B. 60. C. 63. D. 61. 10 pts
2. Why does a t interval use a larger critical multiplier than z at small degrees of freedom?	A. Estimating SD adds uncertainty reflected in heavier tails. B. The population mean is random in the formula. C. The confidence level is automatically 100%. D. The observations must all be larger. 10 pts
3. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The normal model is exact at every tail. B. It is approximation error outside the true support. C. The true sample mean can actually be below every observation. D. The observations must include negative waits. 10 pts
4. Increasing n from 100 to 400 halves SE. Would adding the same 300 observations to any starting n always halve SE?	A. Yes; SE depends only on added observations. B. No; SE never changes. C. No; the ratio of sample sizes matters. D. Yes; an extra 300 always halves it. 10 pts
5. In 1,000 repetitions, each sample contains 25 independent observations. What n belongs in SD/sqrt(n) for the mean?	A. 1,000. B. 25,000. C. 25. D. 975. 10 pts
6. Which continuous event uses the appropriate continuity correction for X>=45?	A. Y>45.5. B. Y>44.5. C. Y=45. D. Y<44.5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **32475**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. Yes; that is the meaning of any confidence interval. B. No; it estimates the population mean. C. No; it instead guarantees every duration lies there. D. Yes; the mean and individuals have identical spread. 10 pts
8. A fixed 5-kg container is added to a shipment of random item masses. What changes?	A. Expected total stays fixed. B. Expected total rises by 5 kg; total SD is unchanged. C. Both mean and SD rise by 5 kg. D. Total SD doubles. 10 pts
9. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 0.25. B. 120. C. 30. D. 90. 10 pts
10. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 12/44. B. 8/44. C. 10/40. D. 10/44. 10 pts

32475

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3.

B.

10 pts

4.

C.

10 pts

5.

C.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9. A.

10 pts

10.

D.

10 pts